



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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UNIT TEST-2, AUGUST - 2024

BUSINESS MATHEMATICS AND STATISTICS

Part - I

Time Allowed : 1.30 Hours]

[Max. Marks : 45

1. Answer all the questions by choosing the correct answer from the given 4 alternatives.
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark.

10x1=10

1. Area bounded by the curve $y = e^{-2x}$ between the limits $0 \leq x \leq \infty$ is
 (1) 1 sq.units (2) $\frac{1}{2}$ sq.unit (3) 5 sq.units (4) 2 sq.units
2. The profit of a function $p(x)$ is maximum when
 (1) $MC - MR = 0$ (2) $MC=0$ (3) $MR=0$ (4) $MC+MR=0$
3. Area bounded by $y = x$ between the lines $y = 1, y = 2$ with $y =$ axis is
 (1) $\frac{1}{2}$ sq.units (2) $\frac{5}{2}$ sq.units (3) $\frac{3}{2}$ sq.units (4) 1 sq.unit
4. Area bounded by $y = |x|$ between the limits 0 and 2 is
 (1) 1 sq.units (2) 3 sq.units (3) 2 sq.units (4) 4 sq.units
5. The order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^{\frac{3}{2}} - \sqrt{\left(\frac{dy}{dx}\right)} - 4 = 0$ are respectively.
 (1) 2 and 6 (2) 3 and 6 (3) 1 and 4 (4) 2 and 4
6. The integrating factor of the differential equation $\frac{dx}{dy} + Px = Q$ is
 (1) $e^{\int Pdx}$ (2) $e^{\int Pdy}$ (3) $\int Pdy$ (4) $e^{\int Pdy}$
7. The P.I of $(3D^2 + D - 14)y = 13e^{2x}$ is
 (1) $\frac{x}{2}e^{2x}$ (2) xe^{2x} (3) $\frac{x^2}{2}e^{2x}$ (4) $13xe^{2x}$
8. The solution of the differential equation $\frac{dy}{dx} = \frac{y}{x} + \frac{f\left(\frac{y}{x}\right)}{f\left(\frac{y}{x}\right)}$ is
 (1) $f\left(\frac{y}{x}\right) = kx$ (2) $x f\left(\frac{y}{x}\right) = k$ (3) $f\left(\frac{y}{x}\right) = ky$ (4) $y f\left(\frac{y}{x}\right) = k$
9. Integrating factor of $\frac{dy}{dx} + \frac{y}{x} = x^3$
 (1) e^x (2) x (3) $\frac{1}{x}$ (4) $\frac{1}{x^2}$
10. The area bounded by the parabola $y^2 = 8x$ bounded by its latus rectum is
 (1) $\frac{16}{3}$ sq.units (2) $\frac{32}{3}$ sq.units (3) $\frac{64}{3}$ sq.units (4) $\frac{4}{3}$ sq.units

PART - II

4 x 2 = 8

1. Answer any 4 questions
2. Each question carries 2 marks
3. Question 16 is compulsory
11. Find the area bounded by $y = 4x + 3$ with x - axis between the lines $x = 1$ and $x = 4$
12. Calculate consumer's surplus if the demand function $p = 50 - 2x$ and $x = 20$
13. If $MR = 20 - 5x + 3x^2$, find total revenue function.
14. Find the differential equation of the family of straight lines $y = mx + c$ when m and c both are arbitrary constants.
15. Solve: $ydx - xdy = 0$
16. Solve $(3D^2 + D - 14)y = 0$



1. Answer any 4 questions

4 x 3 = 12

2. Each question carries 3 marks

3. Question 22 is compulsory

17. Using integration, find the area of the region bounded by the line $y - 1 = x$, the x axis and the ordinates $x = -2$, $x = 3$.

18. Find the revenue function and the demand function if the marginal revenue for x units is $MR = 10 + 3x - x^2$.

19. The demand and supply functions under perfect competition are $p_d = 1600 - 2x^2$ and $p_s = 2x^2 + 400$ respectively. Find the producer's surplus.

20. Solve: $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$

21. Solve the homogeneous differential equation: $x \frac{dy}{dx} = x + y$

22. Solve: $x \frac{dy}{dx} + 2y = x^4$

PART - IV

1. Answer all the questions

3 x 5 = 15

2. Each question carries 5 marks

23. a) Under perfect competition for a commodity the demand and supply laws are $p_d = \frac{8}{x+1} - 2$ and $p_s = \frac{x+3}{2}$ respectively. Find the consumer's and producer's surplus.

(OR)

b) The marginal cost $C'(x)$ and marginal revenue $R'(x)$ are given by $C'(x) = 50 + \frac{x}{50}$ and $R'(x) = 60$. The fixed cost is ₹200. Determine the maximum profit.

24. a) Suppose that the quantity demanded $Q_d = 13 - 6p + 2 \frac{dp}{dt} + \frac{d^2p}{dt^2}$ and quantity supplied $Q_s = -3 + 2p$ where p is the price. Find the equilibrium price for market clearance.

(OR)

b) Solve $\frac{d^2y}{dx^2} + 16y = 0$

25. a) Solve the differential equation $y^2 dx + (xy + x^2) dy = 0$

(OR)

b) Find the area of the region bounded by the curve between the parabola $y = 8x^2 - 4x + 6$ the y -axis and the ordinate at $x = 2$.

